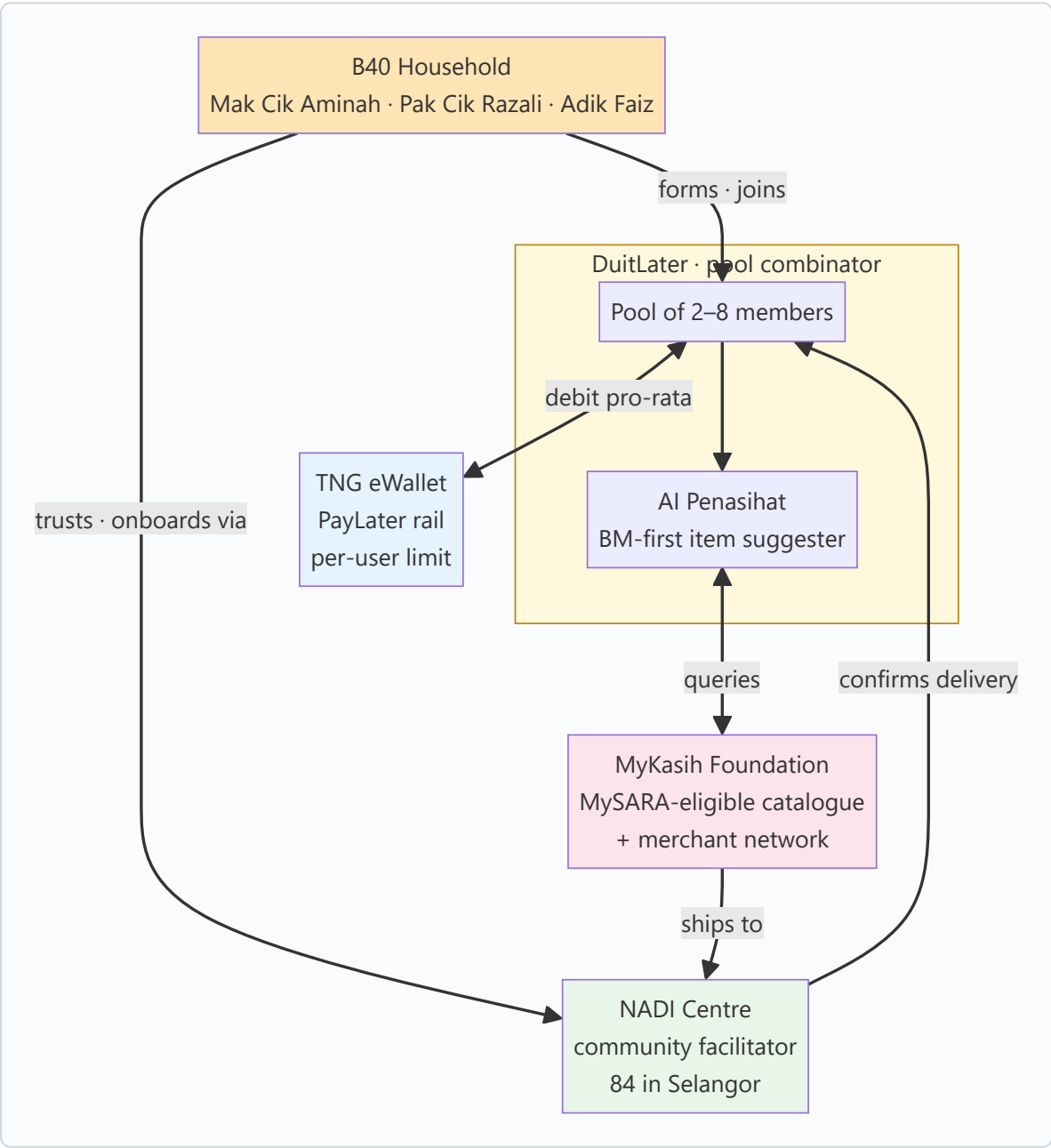


# DuitLater — PRD Visual Flow

**Companion to** `PRD.md` — same product spec, expressed as flow diagrams for fast comprehension during pitch, stand-up, and handoff.

**Source of truth:** [PRD.md](#). If diagrams here disagree with the PRD, the PRD wins.

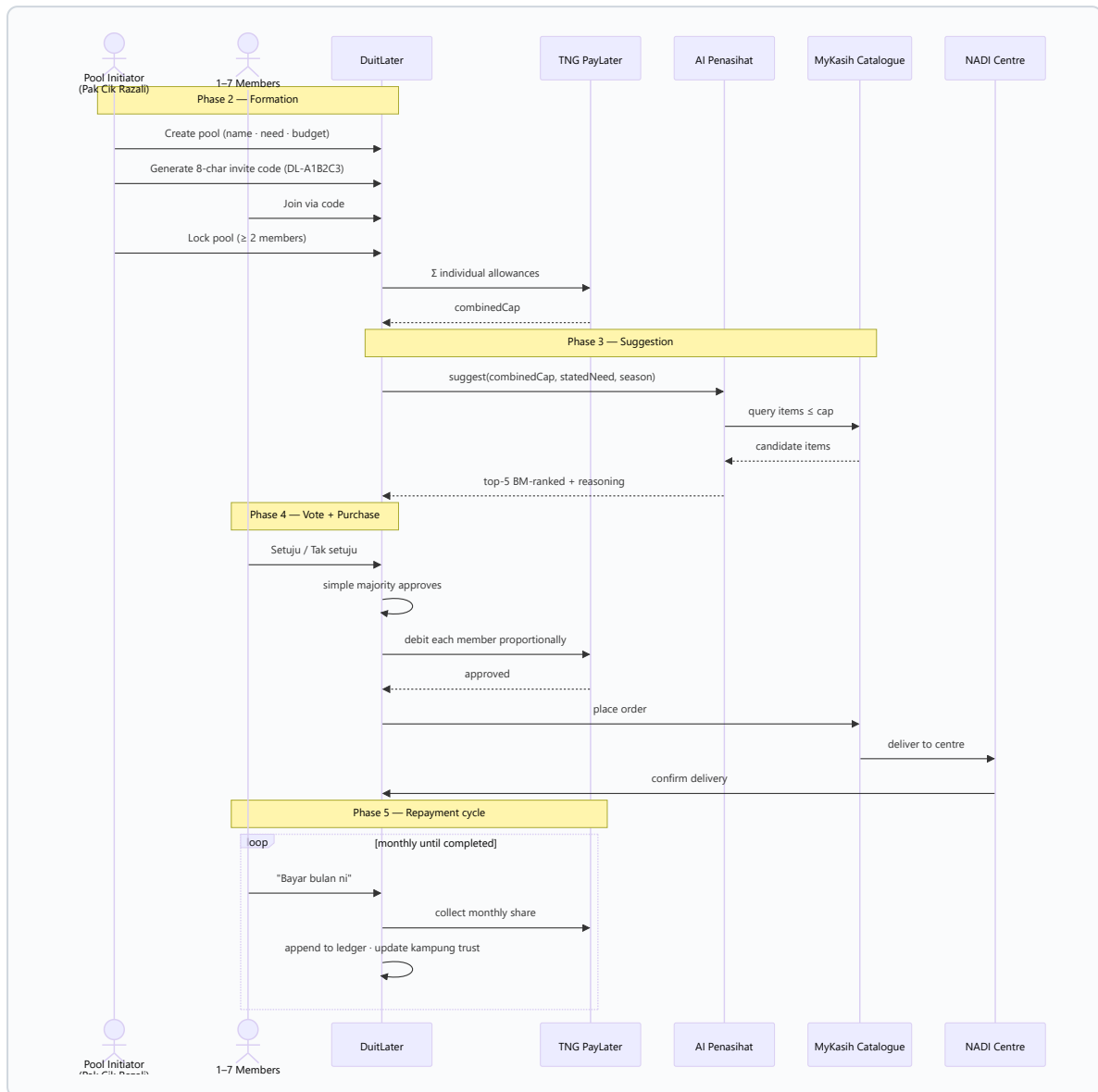
## 1. The Institutional Package (4-way Malaysian)



**Key insight:** zero new institutions invented. Every actor already exists in Malaysia today.  
DuitLater is the *missing connector*.

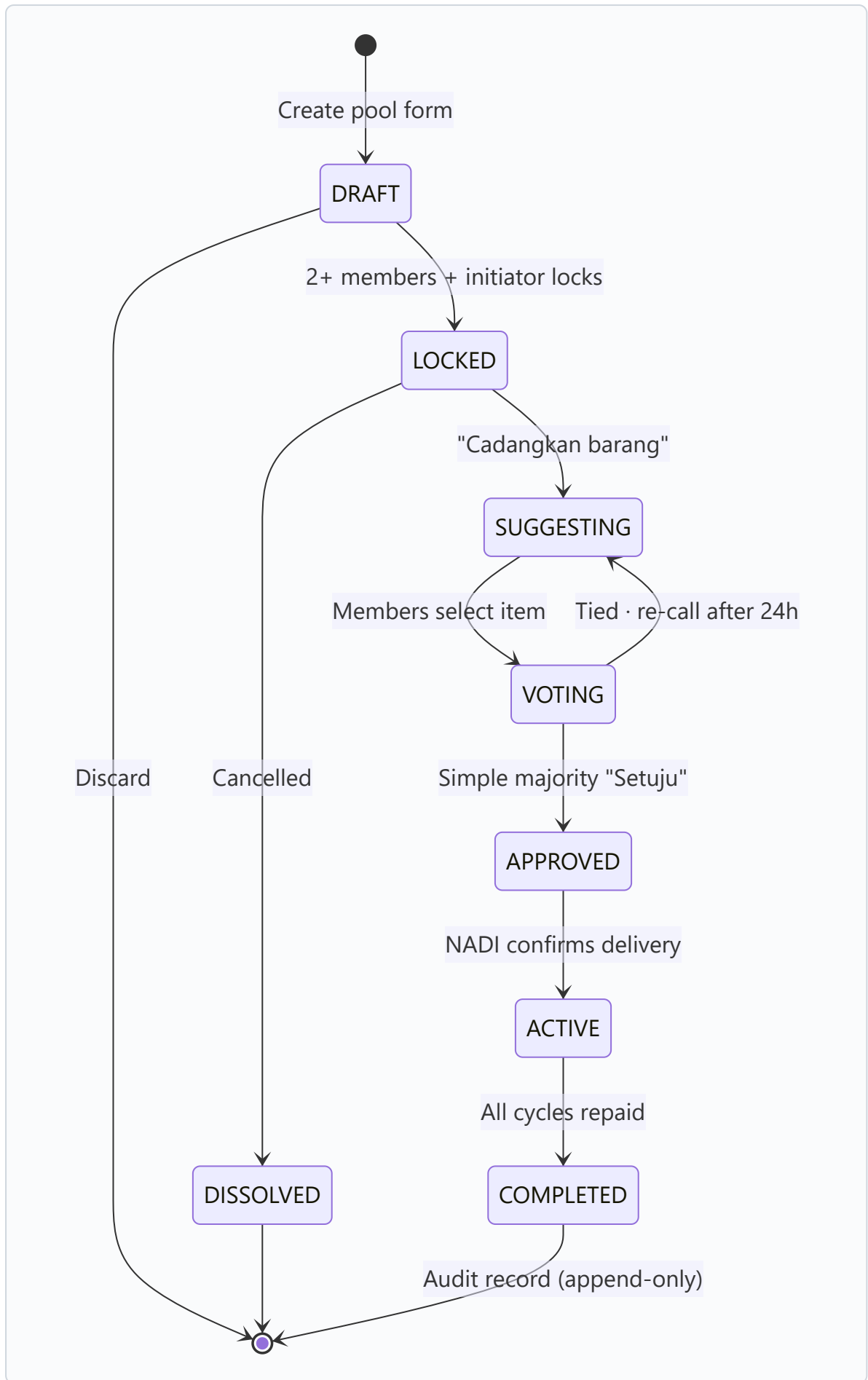
---

## 2. End-to-End User Journey



### 3. Pool Lifecycle (state machine)

---



| State      | Mutations allowed                 | Owner          |
|------------|-----------------------------------|----------------|
| DRAFT      | Edit any field                    | Initiator      |
| LOCKED     | Roster frozen · combinedCap fixed | TNG            |
| SUGGESTING | Penasihat queries Claude API      | AI             |
| VOTING     | Each member votes once            | Members        |
| APPROVED   | TNG debits committed              | TNG + Backend  |
| ACTIVE     | Monthly repayments accrue         | Members + NADI |
| COMPLETED  | Read-only audit                   | None           |

---



## 5. Pool Math — Worked Example

Pool: "Mesin Jahit Felda Gedangsa"

Members + individual TNG PayLater allowances

|                |    |                     |
|----------------|----|---------------------|
| Mak Cik Aminah | RM | 300                 |
| Pak Cik Razali | RM | 400                 |
| Adik Faiz      | RM | 500                 |
| 3 others (avg) | RM | 1,200 (RM 400 each) |

combinedCap: RM 2,400

Item chosen by majority vote: Industrial sewing machine RM 1,800

Proportional debit per member:

|         |   |                          |   |           |
|---------|---|--------------------------|---|-----------|
| Aminah  | → | $300 / 2400 \times 1800$ | = | RM 225.00 |
| Razali  | → | $400 / 2400 \times 1800$ | = | RM 300.00 |
| Faiz    | → | $500 / 2400 \times 1800$ | = | RM 375.00 |
| Other-1 | → | $400 / 2400 \times 1800$ | = | RM 300.00 |
| Other-2 | → | $400 / 2400 \times 1800$ | = | RM 300.00 |
| Other-3 | → | $400 / 2400 \times 1800$ | = | RM 300.00 |

Total committed: RM 1,800.00 ✓

Monthly share over 6 cycles:

|        |   |                       |
|--------|---|-----------------------|
| Aminah | → | RM 37.50 / month      |
| Razali | → | RM 50.00 / month      |
| Faiz   | → | RM 62.50 / month      |
| Others | → | RM 50.00 / month each |

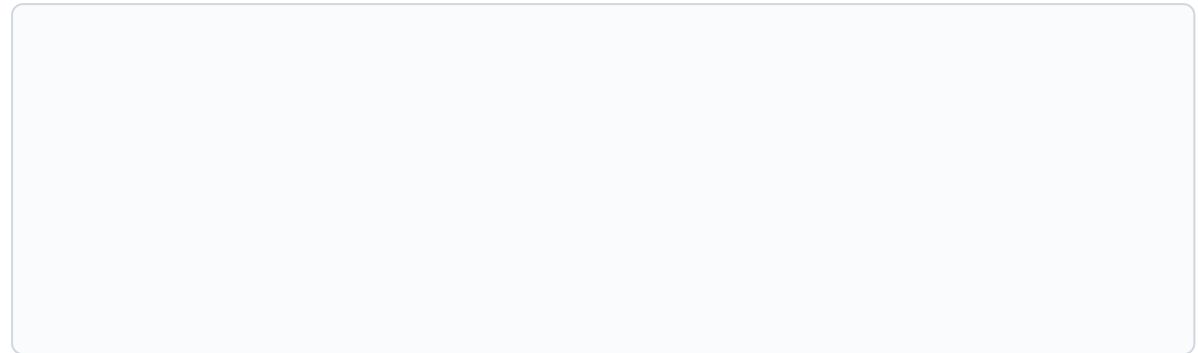
**Invariants** ( PRD \$10 + \$12 ):

- All money in **integer cents** — never float
- Repayment ledger **append-only** — corrections via compensating rows
- Pool.combinedCap **frozen at lock** — never recalculated



## 6. Tech Architecture

---

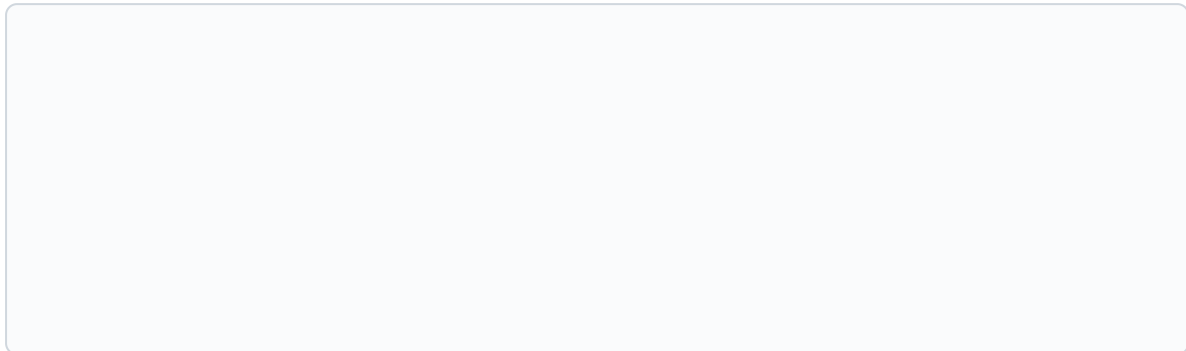


### Repository layout — pnpm workspace monorepo:

```
R2-D2-Finhack/  
├─ packages/  
│   ├─ backend/    Hono · Prisma · Better Auth · zod  
│   ├─ frontend/   Next.js 15 · Tailwind v4 · Jotai · TanStack Query · PWA  
│   └─ db/         Prisma schema · migrations · generated client  
├─ docs/{product,tech,team,process,pitch}/  
├─ maji-core/      protocols · heroes · slash commands  
└─ infra/ scripts/ docker-compose.yml
```

## 7. Build Phases Timeline (48-hour window)

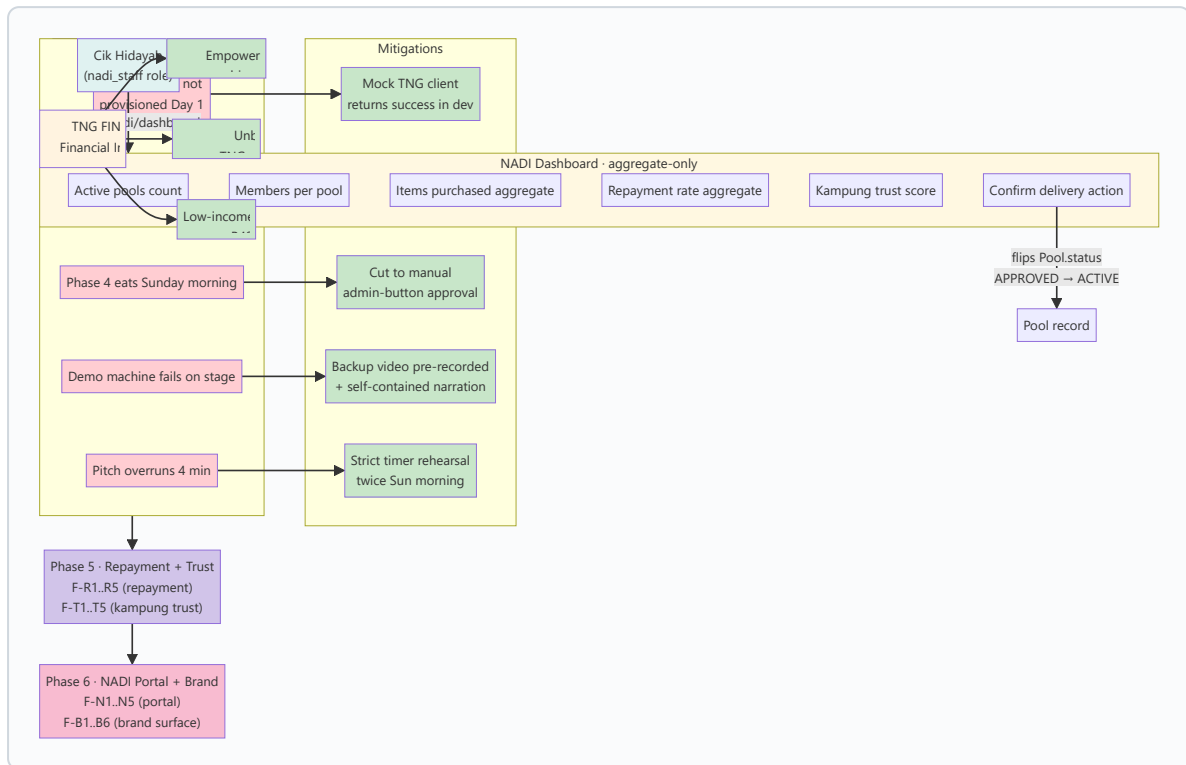
---



| Phase | Lead                      | Testable outcome                                                                    |
|-------|---------------------------|-------------------------------------------------------------------------------------|
| 0     | Mung + Akmal · Kairu gate | <code>docker compose up</code> · <code>/health</code> returns 200 · landing renders |
| 1     | Mung + Akmal              | Register → login → dashboard shows individual PayLater allowance                    |
| 2     | Akmal + Mung              | Create pool → invite → 2nd user joins → lock → combinedCap visible                  |
| 3     | Mung                      | Locked pool → Penasihat returns top-5 BM-ranked items                               |
| 4     | Mung + Kairu              | Members vote → majority approves → simulated TNG debit → ACTIVE                     |
| 5     | Mung + Akmal              | Monthly Bayar → ledger appends → kampung trust score updates                        |
| 6     | Ijam + MatNep             | NADI portal · pitch deck · 4-min on-stage rehearsal                                 |

---

## 8. Functional Requirements Coverage by Phase



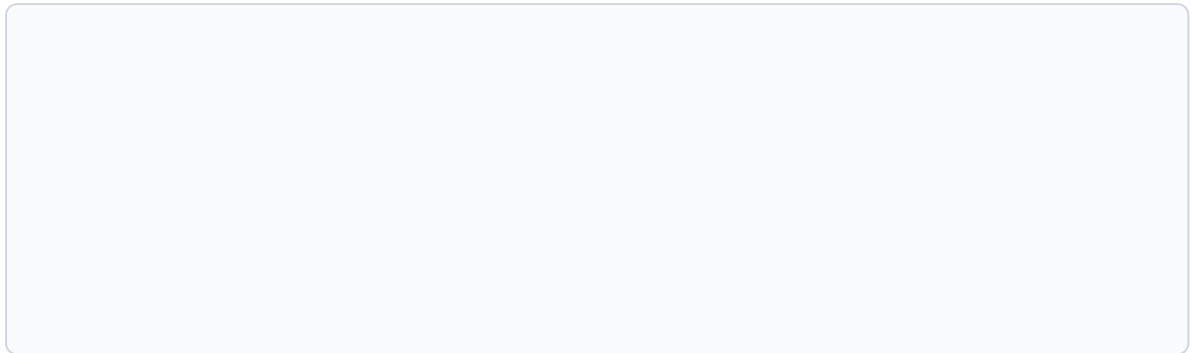
## 9. Risk → Mitigation Map

---

---

## 10. NADI Portal Scope (F-N1..N5)

---

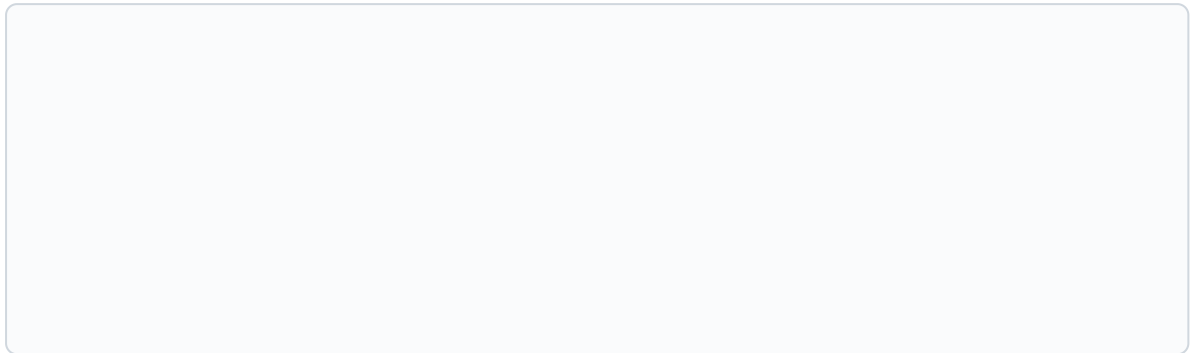


**Privacy guarantee (F-N5):** NADI staff see *aggregate* numbers — never individual member amounts or transactions.

---

## 11. Track-fit Recap (Financial Inclusion)

---



---

*PRD-flow v1.0 · 2026-04-25 · derived from PRD v2.0 · companion visualisation for the same Financial Inclusion track submission.*

*"Sendiri tak mampu, ramai-ramai boleh."*